Supplementary Material for: Flood Prediction When the River Gauge Fails: A Preregistered Evaluation of Physics Priors Built From Weather Data Alone

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S1. Preregistration and provenance mechanics

The main text states that the analysis plan was locked before any confirmatory cell (a prior tested on one basin) was trained, and that two recorded checks support pooling the original and the new seeds. This section gives the full record behind both statements, together with the measured limit on reproducing a stored run.

The hash lock and what the plan predicts

A plan that can be edited after the results arrive is not a plan. Ours was committed to the repository. A second commit that contained nothing but the first commit's cryptographic hash then sealed it. Any later edit to the plan would be visible as a commit that is neither of those two. Every decision rule reported in the paper was fixed in that plan before the cells it judges existed.

The plan is explicit about what it does and does not predict. The original ten seeds predate it and are reused unmodified, so nothing in the plan predicts them. The plan covers only cells that did not exist when it was written: the twenty new seeds per basin and the entire window grid.

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The equivalence bound and the verdict machinery

The bound is derived from the positive arm of the original ten-seed run alone. Here ε is half the smallest confirmed positive effect, where confirmed positive means a basin whose ten-seed paired test has mean effect above zero with $p < 0.05$. Two basins qualify, and the smaller anchor is 01013500 at $+0.041228$. Half of that anchor sets $\varepsilon_{\Delta\mathrm{NSE}} = 0.020614$.

The plan freezes this value for the whole study and forbids recomputing it from the thirty-seed data. A bound recomputed from the same data the equivalence tests judge would be a function of the results it is judging. The bound is used at full precision, since rounding it can flip a verdict whose interval endpoint lands near the boundary. A companion flood-peak bound of 11.3566 % is carried forward for completeness, and it governs no decision, because the flood endpoint is descriptive.

The verdict rules themselves are implemented in the study's equivalence code, which predates the plan and is validated against the taxonomy's worked examples in the unit tests. The plan freezes that machinery whole: no new verdict label, no new threshold, and no change to any boundary inequality.

Seed policy

The confirmatory seed count was fixed at thirty in advance, as the contiguous block 0 through 29, so the seed set is fully determined by the count and carries no selection. There is no interim look, no significance-dependent stopping, and no provision for adding seeds when a result sits near a threshold. If a thirty-seed result is ambiguous, it is reported as ambiguous.

The multiplicity stance in full

Inference is per basin and is reported per basin. The confirmatory experiment runs eleven basins at one endpoint and one mask. Each basin gets its own paired test and its own 90 % interval, with no family-wise claim over the eleven and no correction across them. This is a deliberate stance rather than an oversight, because the question is what happens in each basin, and a family-wise statement over eleven hand-picked, non-exchangeable catchments would not mean much. The stance has a cost, and we accept it: no claim of the form "at least one basin shows an effect" is made anywhere in the paper, because that is exactly the claim per-basin uncorrected testing does not license. Every per-basin effect claim names its basin.

Three claims are deliberately cross-cell, and all three were preregistered as such: the conjunctive specificity aggregation, the refutation branches that

count basins, and the window experiment's outcome label, which is a count over four basins. Where a negative-control cell is ever quoted stand-alone rather than as part of the conjunction, the plan carries forward a rule from the original run: a Bonferroni-adjusted bound at $\alpha = 0.05/4$ accompanies it.

Provenance of the pooled seeds

A frozen protocol is only as strong as the record showing it was followed. Every headline statistic derives from runs whose complete configuration block is embedded in the released result artifact, so the protocol is recoverable from disk alone. Two recorded checks support the pooling of old and new seeds, and each has a stated scope.

The first is a comparison captured before this study's second stage touched any training code. It shows that no training-path source file changed between the archived state that produced the original ten seeds and the state that trained the new twenty. The second is an assertion, made by the artifact builder, that all 660 per-run configuration blocks agree. That check is narrower than it sounds, since it covers only the eight configuration scalars the blocks record and cannot see the code path, the splits, or the architecture. The recorded code comparison is what covers those.

The reproducibility limit

Cross-session reproducibility has a measured limit, and we state it rather than assume determinism. Rerunning one stored cell in the current environment does not reproduce its result file bit for bit. The physics arm differs by at most 1.6×10^{-5} in relative terms across all reported metrics, and the control arm reproduces exactly. At the endpoint the drift is 1.8×10^{-6}, roughly one part in 100,000 of that basin's measured effect and three to four orders of magnitude below its interval half-width. We tested the obvious explanation, that thread count changes the order of floating-point summation, by rerunning with thread counts pinned to one.

S2. Training-protocol detail

The main text summarizes the frozen training protocol in a table and names each gauge only in the per-basin results table. This section gives the study roster with its regime labels, the data preparation, the masking and interval mechanics, the seed-pooling rules, the constants of the priors, and the parts of the physics implementation that no reported result uses.

Study roster

The study uses 11 CAMELS-US basins carrying 7 distinct regime labels: snowmelt-montane (×3), humid rain (×2), arid/flashy (×2), mixed rain-snow, groundwater/baseflow, energy-limited maritime, and Mediterranean. The 11 gauges are 11266500, 09352900, 09210500, 01013500, 01466500, 07067000, 12040500, 02177000, 11143000, 08202700, and 09512280. The four gauges the snow gate passes have long-term snow fractions from 0.298 to 0.871, recomputed from the training segment with the production code for this paper.

Data preparation

For each basin we use the CAMELS-US daily basin-mean forcing series (precipitation, maximum and minimum temperature, solar radiation, and vapour pressure) and the CAMELS-US daily discharge series with quality flags. The meteorological drivers are treated as always-observed inputs. Only the discharge channel is subject to the simulated sensor failure. All normalizations are per-feature standard scaling, and the data-derived constants fitted on the training segment are the loss thresholds and the master recession constant. Days with the missing-discharge sentinel are linearly interpolated before windowing. Daily mean temperature for the snow model is the mean of the daily maximum and minimum.

Loss and masking mechanics

The NSE loss is the mean squared error normalized by the (variance-stabilized) target variance, so minimizing it maximizes Nash–Sutcliffe efficiency. Both arms train on this same loss, which is the mean squared prediction error divided by the training-segment discharge variance.

A masked day is represented by zeroing the normalized discharge input and setting a companion binary observed-flag channel to 0. The flag channel is what lets the network distinguish a missing reading from a genuine average-flow day, since 0 in normalized space equals the training-mean discharge.

At training time the masked fraction of each window is random, drawn uniformly on [0, 1.0] and seeded. At test time masks are drawn at the fixed evaluation levels 0 %, 80 %, and 100 % from a dedicated evaluation-mask seed common to all runs.

Seed pooling and the frozen constants

The protocol was defined and first run with ten paired seeds (0–9) per basin and arm. The confirmatory experiment reported in the paper extends it to thirty paired seeds (0–29): it adds seeds 10 through 29 for all eleven basins in both arms and pools all thirty into one test per basin. The window-length experiment runs its own independent cells at ten seeds.

Every constant of the protocol other than the seed count and the input-window length is frozen across both experiments: the architecture, the optimizer settings, the epoch budget and early stopping, the masking regime, the splits and their date boundaries, the train-only fitting of every data-derived constant, the snow-gate membership, and the loss. If any frozen constant had changed between the original run and the extension, the affected basin's thirty-seed test would be invalid, and the plan's rule is to rerun the original seeds rather than pool across a change.

Interval arithmetic and reused control arms

The reported metrics are NSE, MAE (mean absolute error), RMSE (root-mean-square error), KGE (Kling–Gupta efficiency, a complementary skill score), and flood-peak MAE. The last is defined as the MAE over test days whose observed discharge is at or above the 95th percentile of the evaluated test series.

The interval is the mean paired difference plus or minus the Student-t quantile times the standard error of the differences, with the sample standard deviation on $n - 1$. The quantile is taken at 29 degrees of freedom for the thirty-seed confirmatory cells and at 9 for the ten-seed window cells. The control arm does not depend on which prior is enabled, so the negative-control experiments reuse the frozen protocol's control arms rather than retraining them. The configuration resolution guarantees the reused cells are the same runs.

Prior integration and the unused loss-side alternative

Each active prior contributes its state series as extra input channels: SWE (snow water equivalent) and w^{eff} (the effective water input) for the snowpack, the antecedent-precipitation index API for the antecedent-moisture (AMC) bucket, and S for the recession store. In the physics arm these are normalized with training-segment statistics and appended to the seven base channels. The physics therefore enters the evaluated models through the input channels alone, not through the loss.

The released code also carries a loss-side alternative as an opt-in switch. It is a physics-consistency penalty, the mean of $(\hat{q} - r)^2$ times a fixed weight

of 0.1. The penalty pulls the prediction $\hat{q}$ toward a causal expected-release reference r. That reference is an exponential moving average of w^{eff} with decay 0.9. A diagnostic round found the penalty net-unhelpful, and it is off in every run the paper reports.

Snowpack constants

The degree-day snow model's three constants are set in advance and are never fitted to data: rain/snow threshold $T_s = 1.0$ °C, melt threshold $T_m = 0.0$ °C, and degree-day melt factor $d = 2.5$ mm °C^{-1} d^{-1}.

Recession constant: definition and full-record fits

The master recession constant k_b is the median falling-limb ratio Q_{t+1}/Q_t above a low-flow threshold, which pins the store's dry-day decay to the basin's own drainage rate.

The train-only fits on the two target storage basins are k = 0.9444 and 0.9333, e-folding times 17.5 and 14.5 d. Fitting the same two basins on the full record instead gives k = 0.9474 (18.5 d) and 0.9333 (14.5 d). The reported runs use the train-only fits.

Run count and test windows in the window-length experiment

The window grid adds 330 training runs. The usable test set shrinks as the window grows, because a model with a longer window can form fewer complete windows over a fixed test segment. The counts are 1080, 1065, 1035, and 1005 test windows at 15, 30, 60, and 90 days. We report the per-column count with every result rather than assume the difference away.

S3. Deviations and post-data decisions

The main text names six departures from the preregistered plan and summarizes the ones that qualify a reported claim. This section is the full ledger, in the same order.

Preregistration only earns trust if departures from it are reported. This study has six, and we state them plainly.

First, the rule that assigns the window experiment's outcome label was corrected after the window data existed. The analyzer was written and committed before any window cell was trained. Its first version decided the label from the four-point ordering count, which the plan designates a descriptive summary, while the plan names the paired contrast as the confirmatory quantity. A statistical audit caught the contradiction once the grid was complete, and the rule was corrected in a later, identified commit so that

the label is computed from the contrast alone. The corrected function no longer receives the ordering data at all. Two facts bound what the timing could have bought. The confirmatory quantity was specified before the data, so moving the label onto it is largely conformance. And the label is *partial* under both the original and corrected rules, so the headline does not move. What changed is the stated reason, from an ordering count of one in four to a contrast count of two in four.

Second, the threshold that separates a supported outcome from a partial one, namely that all four snow basins confirm, appears nowhere in the plan. It was chosen in the same post-data commit that corrected the outcome rule. It is the strictest threshold available, so it cannot have been selected to flatter the result. Even so, calling the correction pure conformance would overstate it: the confirmatory quantity was preregistered and the threshold was not.

Third, the plan's cross-basin span-ordering prediction was evaluated late. It was computed after the pre-data analyzer, in the same round that produced the findings the paper reports, and we state that plainly for the same reason we disclose the outcome-rule timing. Results reports its outcome: the prediction failed outright.

Fourth, we ruled that the stand-alone Bonferroni provision carries over to the thirty-seed negative-control cells, and we made that ruling after seeing the data. It changes nothing. The AMC cells sit at p near 10^{-15} and the recession cells at 0.185 and 0.294, all far from the adjusted level in one direction or the other. The correction's real analogue for an equivalence verdict is a wider interval, and wider intervals can move a verdict. The nearest cell survives that too. The recession cell on 11143000, judged at a Bonferroni-consistent level, has an interval roughly half again as wide and still sits inside the bound, though with little room to spare.

Fifth, every confirmatory interval in this study is a 90 % interval, which corresponds to a two-sided α of 0.10, twice the conventional rate. No correction was applied across the window experiment's six contrasts. The confirming count is sensitive to that choice: the contrast on 11266500 confirms with $\mathrm{p} = 6.627 \times 10^{-2}$ and would not confirm at $\alpha = 0.05$, which would take the window experiment's headline from two basins of four to one. The count is not sensitive to the choice of test. An exact two-sided Wilcoxon signed-rank test on the same six per-seed contrasts moves no contrast across the 0.10 boundary, and 11266500 again confirms narrowly, at $\mathrm{p} = 8.398 \times 10^{-2}$. The reader should have the level and the absence of correction stated rather than left to be inferred.

Sixth, the split-half comparison of the original ten seeds against the

twenty new ones was not preregistered. It was run after the thirty-seed data existed, to check whether tripling the seed budget changed any basin's sign or significance status. It is computed directly from the pooled result file rather than from a stored analysis artifact, the one claim in this paper without one.

The reported pipeline supersedes an earlier internal version, in which a recession-feature implementation error was found and corrected before any reported result was produced. This correction was routine pre-submission quality assurance, not a deviation from the preregistered plan.

S4. State-memory measurement

The main text defines the memory measurement in one sentence and reports its analytic reference values. This section gives the estimator, the input and release series for each prior, the internal checks, and the alternative measure that was computed and then rejected.

For each prior × basin cell, we measure how long the reconstructed hidden state retains forcing information and compare that memory with the model's input window.

Primary estimator: the amount-weighted centroid lag

Per complete water year (Oct 1 – Sep 30, labeled by ending year), the lag is the difference between the amount-weighted mean date of release from the store and the amount-weighted mean date of input to it:

$$\text{lag} \;=\; \frac{\sum_t t\,\text{release}_t}{\sum_t \text{release}_t} \;-\; \frac{\sum_t t\,\text{input}_t}{\sum_t \text{input}_t} \tag{S1}$$

with t = days since Oct 1. This is the mean residence time of a unit of water in the store. It is a causal input-to-release delay, and it directly operationalizes how long forcing information is stored before it can affect discharge. Water years with total input below 10 mm are treated as null. Results are reported as mean ± standard deviation across complete water years of the study window (up to 19 per basin), and those spreads are substantial relative to the means, which Results states alongside every use.

For the snow store, input is accumulation-day snowfall (the model's own rain/snow branching) and release is the model's melt output. For the AMC bucket, input is total precipitation, because all precipitation enters the composite precipitation → snow store → bucket path whose terminal state is measured. Release is the bucket's daily decay leak $(1 - k) \cdot \text{API}_{t-1}$.

Mass balance over the full series is asserted at run time, and a cascade decomposition attributes the composite memory to the snow path versus the bucket, since centroid lags are additive along a cascade. All series come from the shipped production loader with all defaults, and an internal check confirms that two independent recomputations of each state from raw forcing produce identical arrays. Parameter identity with the training pipeline is therefore structural rather than assumed.

The analytic reference memories of the two filter priors follow directly from their decay constants: e-folding $-1/\ln(\mathrm{k}) = 13.8$ d for the AMC bucket ($\mathrm{k} = 0.93$) and 17.5 and 14.5 d for the train-fitted recession filters ($\mathrm{k} = 0.9444$ and 0.9333).

Rejected alternative: autocorrelation e-folding

An autocorrelation-based memory is computed for comparability but is inadmissible as the primary metric. It is the first lag at which the autocorrelation of the mean-removed daily state drops below 1/e. The deterministic seasonal cycle inflates the autocorrelation of any winter-peaked variable, so a long e-fold would partly restate that winter is long rather than measure information storage. The transfer lag isolates the input $\rightarrow$ release delay that the input window must span. The autocorrelation reading corroborates but carries no claim.

Provenance of the memory figures

These measurements are carried unchanged from the original run and were not recomputed for the window experiment. They are fixed, parameter-free calculations on public data with no training component, which is why they carry no seed uncertainty. The window experiment's predictions take them as inputs.

S5. Per-cell results detail

The main text reports the frozen-protocol and window-experiment results at the level of verdicts, headline effects, and caveats. This section carries the per-cell numeric detail behind them: the seed-split arithmetic, the mask-level and flood-peak values, the run-configuration record behind the seven zeros, the per-window profiles and verdict bounds, and the memory-to-window ratios.

Seed-split detail

Tripling the seed budget left the negative pair in place. On 09352900 the thirty-seed effect is essentially unchanged from its ten-seed estimate of −0.129. On 09210500 it moved from −0.099 to −0.074, and the twenty new seeds on their own give −0.061. That gap between the original ten-seed batch and the twenty new seeds is within sampling variation for batches of this size. The estimate nonetheless stays far from zero, and its interval stays clear of it. The split-half check's second criterion, no change in significance status, is weak by construction. The weakest per-half evidence among the four basins is $p = 3.5 \times 10^{-3}$, on 01013500's original ten seeds, so no basin's significance status could have changed.

Mask-level and flood-peak values

With no discharge missing, the physics arm's ΔNSE is −0.070 on 11266500, −0.015 on 09352900, and −0.018 on 01013500. On 09210500 the interval spans zero (+0.006, −0.005 to +0.017). With 80 % of the record missing, all four gated basins are negative.

Under total discharge loss, flood-peak error falls by 51.3 % on 11266500 (90 % CI 49.9 to 52.7) and by 23.0 % on 01013500 (19.7 to 26.4). It rises on the other two basins: the change is −45.4 % on 09352900 (−49.2 to −41.6) and −2.5 % on 09210500 (−5.0 to −0.1). Both endpoints are descriptive under the plan and decide nothing.

The run-configuration record behind the seven zeros

On a non-gated basin the physics arm records the snow feature off and an input of seven channels, identical to the control arm. On a gated basin it records the feature on, with nine. The two arms on a non-gated basin are therefore the same code path on the same seed, and they produce bit-identical predictions.

Per-window profiles

From 30 days out to 90 the effect declines on 11266500 (+0.166 to +0.130 to +0.097) and on 01013500 (+0.041 to +0.032 to +0.019), and it moves further negative on 09352900 (−0.129 to −0.140 to −0.152). At the short end, the effect at 15 days against 30 days is +0.129 against +0.166 on 11266500 and +0.022 against +0.041 on 01013500, so the benefit peaks at the 30-day window in both helped basins. In the two negative basins a smaller window shrinks the harm, from −0.129 to −0.112 on 09352900 and from −0.099 to −0.050 on 09210500. The full window profile on 09210500

is −0.050, −0.099, −0.051, and −0.074 at 15, 30, 60, and 90 days. It moves against both halves of the reading rather than merely failing to support them. That basin's contrast has a standard deviation of 0.178, against roughly 0.049 for each of the other three snow cells, so its interval is about four times wider than any other snow basin's.

The AMC control on 01013500 is roughly flat in window length, with a 15−90 contrast of +0.014.

Per-window AMC verdict bounds

Judged against the frozen ε at every window, the AMC cell on 01013500 clears the bound at all four windows, with interval lower bounds +0.048, +0.055, +0.033, and +0.028 at 15, 30, 60, and 90 days. On 07067000 it clears the bound at 30, 60, and 90 days (+0.056, +0.059, and +0.069). The exception is 07067000 at 15 days, where the lower bound of +0.016 falls below ε and the verdict is *inconclusive*.

Memory-to-window ratios

The snow-store lags are 4.1 to 6.6 times the 15-day window and 0.69 to 1.09 times the 90-day window, so the tested grid brackets each snow basin's memory from well inside to roughly its own scale. Only one cell in the whole window grid places an AMC state outside its window: AMC on 01013500 at 15 days, at a memory-to-window ratio of 1.44. Every snow basin at that window sits at 4.1 to 6.6. The memory figures were measured once and not recomputed for the window experiment.